

Week 9: Minimum Spanning Trees

A minimum spanning tree connects every vertex of a weighted graph with the least total edge weight.

Algorithms

- Kruskal's algorithm sorts edges by weight.
- Prim's algorithm grows a single tree.
- Borůvka's algorithm merges components in rounds.

Running times

Algorithm	Time
Kruskal	$O(E \log E)$
Prim	$O(E \log V)$

Week 10: Shortest Paths

Dijkstra's algorithm finds shortest paths when all edge weights are non-negative.

Bellman–Ford handles negative weights and detects negative cycles.